QING YAN

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Research Interests

Agentic systems that let robots learn, reason, and act autonomously in human environments: LLM-driven task planning and skill composition for long-horizon manipulation, enhancing pre-trained robot policies with vision-language models and reinforcement learning, and human-robot interaction.

Education

University of California, Los Angeles Sep 2026 – Present
Ph.D. in Electrical & Computer Engineering · Los Angeles, CA
Lin Yang's Research Group, advised by Prof. Lin F. Yang

Johns Hopkins University Aug 2026
M.S.E. in Robotics · Baltimore, MD
Intuitive Computing Lab (LCSR), advised by Prof. Chien-Ming Huang

University of Washington Jun 2024
B.S. in Electrical & Computer Engineering · Seattle, WA
Sensors, Energy & Automation Laboratory (SEAL), Prof. Alexander V. Mamishev

Publications

- Gao, M., **Yan, Q.**, & Li, X. (2026). GUIDES: Guidance Using Instructor-Distilled Embeddings for Pre-trained Robot Policy Enhancement. *IEEE International Conference on Robotics and Automation (ICRA)*, 2026. arxiv.org/abs/2511.03400. Advised by Prof. Chien-Ming Huang (JHU) and Prof. Jiachen Li (UC Riverside).
- Yang, W., & **Yan, Q.** (2026). Scouting by Reward: VLM-to-IRL-Driven Player Selection for Esports. *MIT Sloan Sports Analytics Conference* (under review). arxiv.org/abs/2604.14474. Advised by Prof. Anton Dahbura (JHU).

Research Experience

Lin Yang's Research Group, UCLA Sep 2026 – Present
Ph.D. Researcher · Advised by Prof. Lin F. Yang · Los Angeles, CA

- **Agentic Robot Harness (Skill Composition in RSI):** building an agentic harness for long-horizon manipulation in which an LLM composes task graphs from a skill library and a classical motion planner executes them; studying skill composition and abstraction so the system can expand its own skill set over time.
- **LLM Infrastructure RSI Agent:** developing an agent that iteratively improves LLM infrastructure code through recursive self-improvement.

Intuitive Computing Lab (LCSR), Johns Hopkins University Sep 2024 – Aug 2026
Graduate Researcher, Human-Robot Interaction · Advised by Prof. Chien-Ming Huang · Baltimore, MD

- Developed a framework using VLM semantic guidance to upgrade pre-trained robot policies (e.g., behavior cloning); validated on a physical UR5 arm. Published at ICRA 2026 (GUIDES).
- Built WeakRobotRestaurant, a 2D human-robot interaction game in which limited-capability service robots ask people for help; logged contextual factors (workload, urgency, effort) and applied causal inference to isolate the effect of persuasion strategies on task acceptance.
- Framed the learned context-to-outcome mapping as a proto-mental world model for social delegation, specifying how an embodied robot should represent workload and urgency when choosing persuasion strategies.
- Designed the interactive infrastructure and an LLM-based voice command system for the Stretch RE1 mobile manipulator.

Sports Analytics Research Group, Johns Hopkins University Sep 2025 – Jan 2026
Independent Researcher · Advised by Prof. Anton Dahbura · Baltimore, MD

- Engineered an esports scouting system combining VLM and metadata embeddings with inverse reinforcement learning to identify professional players most similar to a target profile; validated with the esports organization Fnatic.

Sensors, Energy & Automation Laboratory (SEAL), University of Washington Apr 2023 – Apr 2024
Embedded Team Associate · Prof. Alexander V. Mamishev · Seattle, WA

- Designed the signal-receiving module (transformer, preamplifier, DAQ) for the Heimdall ACU antenna system to detect

weak electromagnetic interference.
- Contributed to PCB simulation and noise-reduction experiments; supported the Fruit Ripeness Sensor, Ammonia Sensor, and Weak Signal Instrument projects.

Industry Experience

KeploreAI May 2025 – Present
Cofounder & COO · Santa Clara, CA

- Cofounded the company and its flagship product Kepilot, an AI-driven coding tool built on OpenDevin with environment isolation and Docker integration for academic research and industrial R&D.
- Secured Kepilot deployments and subscriptions across research labs at Stanford, MIT, UC Berkeley, and UCLA.
- Introduced Physmind, a second-generation skills library providing end-to-end R&D solutions for magnetic-material, automotive, and robotics partners (e.g., Tianji Kirin, Geely Auto).
- Closed pre-seed funding and opened an active seed round.

Jianghuai Advance Technology Center (Qiyuan Lab) Jul 2024 – Aug 2024
Control Algorithm Intern · Hefei, China

- Trained the Qijiang No. 1 humanoid robot for straight-leg walking with PPO reinforcement learning, achieving fluent distance walking in simulation.
- Implemented and tuned a Universal Virtual Component (UVC) controller for stable bent-leg walking, moving the control strategy from manual tuning to learning-based methods.

iFlytek Co., Ltd. Jun 2022 – Sep 2022
Technology Department Intern · Hefei, China

- Analyzed intelligent-manufacturing needs across Anhui Province through enterprise surveys, requirement analysis, and statistical reporting, focused on control systems and robotics.

Selected Projects

Machine Learning Competition Agent · IOAI 2026 2026

- Built an agentic framework that autonomously solves machine-learning competition tasks end to end.

Autonomous Self-Driving Wheelchair · ENGINE Capstone, UW 2023 – 2024

- Built an autonomous electric wheelchair (sponsored by Cyberwork Robotics) using a RealSense camera, RPLidar, and IMU. Combined SLAM and YOLO with ROS for navigation and obstacle avoidance, integrated the partner's Skylane lane-keeping system, and added an OGMD module to detect abnormal occupancy-grid features and switch navigation modes.

Technical Skills

Programming: Python, C++, Java, RobotC, Swift
AI & Algorithms: ROS, Computer Vision, Machine Learning / Reinforcement Learning, LLM / VLM agents, SLAM, Causal Inference, ReAct, OpenHands
Hardware & Platforms: Humanoid robots, Stretch RE1, XLeRobot, UR5 arm, autonomous wheelchair, LiDAR, Intel RealSense, PCB design, Arduino, Raspberry Pi, FPGA

Honors & Awards

- Dean's List, University of Washington (every quarter)
- Semifinalist, VEX Robotics World Championship (2019)
- Gold Award, VEX Robotics China National Championship (2019)
- Gold Award, VEX Robotics Asian Open (2018)
- First Prize (×2), China Adolescent Robotics Competition